

Facebook Network Analysis Report using Gephi

1. Introduction

Social network analysis helps in understanding relationships and interactions between users in a network. In this task, a Facebook network dataset was analyzed using Gephi. The network was imported as an undirected graph, visualized using graph layouts, and analyzed using different network analysis techniques such as degree centrality, betweenness centrality, modularity and filtering.

The main objective of this analysis was to identify highly connected nodes, bridge nodes and communities within the Facebook network.

2. Dataset Description

The dataset used in this task represents Facebook friendships.

- Nodes represent Facebook users.
- Edges represent friendship connections between users.
- The graph was imported as an undirected network because friendships are mutual.

3. Network Overview

The imported network contained multiple interconnected users and friendship relationships.

Observations

- Nodes represent users in the Facebook network.
- Edges represent friendship links.
- The network appeared highly dense and clustered.

4. Degree Centrality Analysis

Degree centrality measures the number of direct connections a node has.

Calculation

The Average Degree statistic was executed from the Statistics panel.

Observations

- Nodes with high degree values were highly connected users.
- These nodes acted as influential users in the network.

5. Top 5 Most Connected Nodes

The degree column in Data Laboratory was sorted in descending order to identify the most connected nodes.

Observations

The top nodes had significantly higher connections compared to other nodes in the network.

These nodes represented major hubs in the Facebook network.

6. Betweenness Centrality Analysis

Betweenness centrality identifies nodes that act as bridges between communities.

Calculation

The Network Diameter statistic was executed.

Observations

- Nodes with high betweenness centrality connected different groups.
- These nodes played an important role in communication flow.

7. Identification of Bridge Nodes

Nodes with high betweenness values were identified as bridge nodes.

Importance of Bridge Nodes

- Connect separate communities.
- Help information travel across the network.
- Their removal may disconnect communities.

8. Comparison of Original and Filtered Network

Original Network

- Large and dense.
- Included many weakly connected nodes.
- Harder to interpret visually.

Filtered Network

- Cleaner and more readable.
- Highlighted important nodes and communities.
- Reduced visual clutter.

Observation

Filtering improved clarity and emphasized the core structure of the Facebook network.

9. Conclusion

The Facebook network was successfully analyzed using Gephi. Different network analysis techniques such as degree centrality, betweenness centrality, modularity, and filtering were applied to understand the structure of the network.

The analysis identified:

- Highly connected users,
- Bridge nodes connecting communities,
- Different community structures within the network.

The filtered visualization provided a clearer understanding of the core relationships and important nodes in the Facebook network.